

CS1 Week 10

Annotated

Polar Plots, Nyquist



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Recap Last Week

Quiz 1

Why do we look at the frequency response?

- A.** to visualize the frequency response of the system
- B.** to impose constraints on magnitude and phase for stability
- C.** to synthesize controllers such that the system meets certain specifications
- D.** to look at the system response over many different frequencies ω

Quiz 1

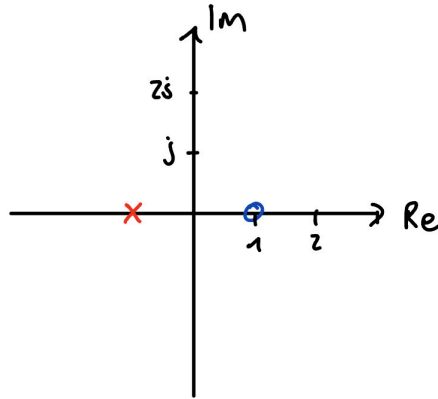
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- A.** to visualize the frequency response of the system
- B.** to impose constraints on magnitude and phase for stability
- C.** to synthesize controllers such that the system meets certain specifications
- D.** to look at the system response over many different frequencies ω

$$u(t) = \cos(\omega t) \xrightarrow{\quad} \boxed{G(j\omega)} \xrightarrow{\quad} y_{ss} = |G(j\omega)| \cos(\omega t + \angle G(j\omega))$$

Recap Frequency Response

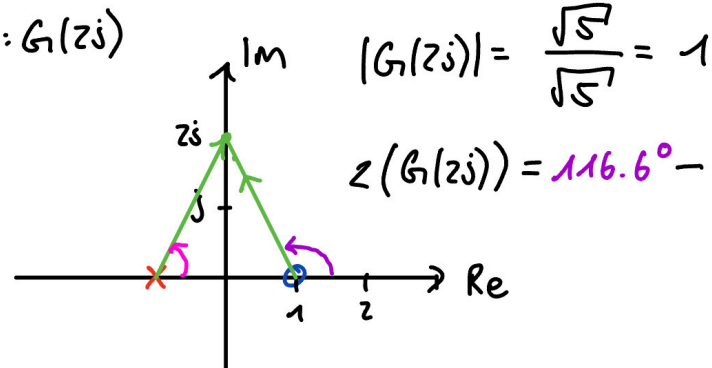
- The magnitude is scaled by $|G(j\omega)|$ and the phase is shifted by $\angle G(j\omega)$
- Frequency ω stays the same
- $|s - p|$: length of vector from p to s
- $\angle(s - p)$: phase from p to s



As you know, we can do this *graphically*

Example: $G(s) = \frac{s-1}{s+1} \rightarrow z_1 = 1, p_1 = -1$

$\omega=2: G(2j)$



$$|G(2j)| = \frac{\sqrt{5}}{\sqrt{5}} = 1$$

$$\angle(G(2j)) = 116.6^\circ - 63.4^\circ = 53.1^\circ$$

Quiz 2

Which of the following statements about Bode plots are correct?

A. We use logarithmic scales for ω and $|G(j\omega)|$

B. $|G_1(j\omega)| \cdot |G_2(j\omega)| = |G_1(j\omega)|_{dB} \cdot |G_2(j\omega)|_{dB}$

C. we can add Bode plots of different TFs on top of each other

D. the magnitude plot is influenced by the sign of k_{bode}

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C. we can add Bode plots of different TFs on top of each other

D. the magnitude plot is influenced by the sign of k_{bode}

Bode Plots

$$u(t) = \cos(\omega t) \rightarrow \boxed{G(j\omega)} \rightarrow y_{ss} = |G(j\omega)| \cos(\omega t + \angle G(j\omega))$$

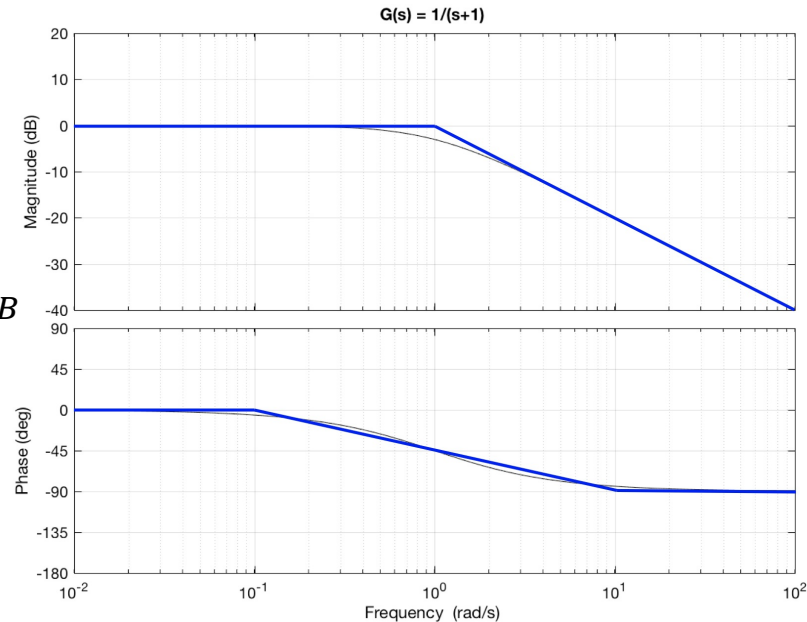
- Horizontal Axis: Frequency ω plotted on a **$\log_{10}$ scale**
- Vertical Axis:
 - Magnitude $|G(j\omega)|$ plotted in decibels (dB)
 - Phase plot $\angle G(j\omega)$ in degrees

$$|G(j\omega)|_{dB} = 20 \cdot \log_{10} |G(j\omega)|$$

$$|G_1(j\omega)| \cdot |G_2(j\omega)| = |G_1(j\omega)|_{dB} + |G_2(j\omega)|_{dB}$$

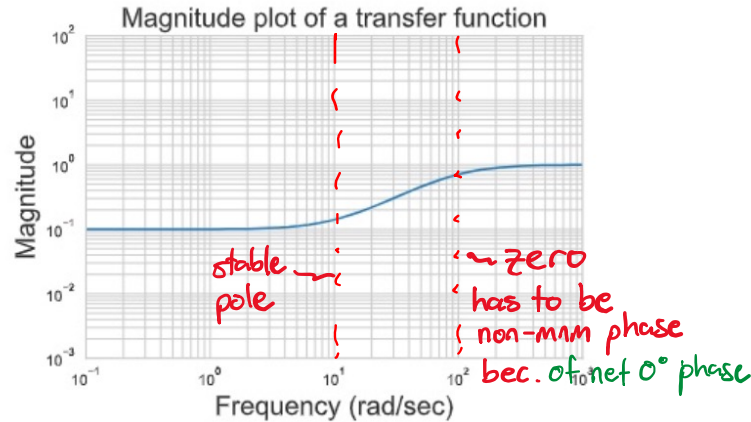
$$\angle G_1(j\omega) G_2(j\omega) = \angle G_1(j\omega) + \angle G_2(j\omega)$$

- Inverting a TF is equivalent to a reflection about the horizontal axis
$$|G(j\omega)^{-1}|_{dB} = -|G(j\omega)|_{dB}$$

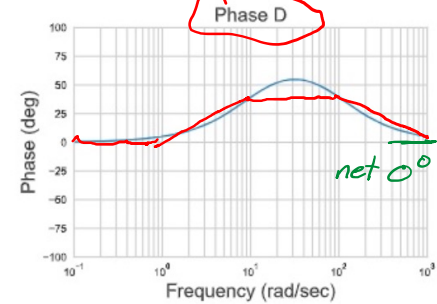
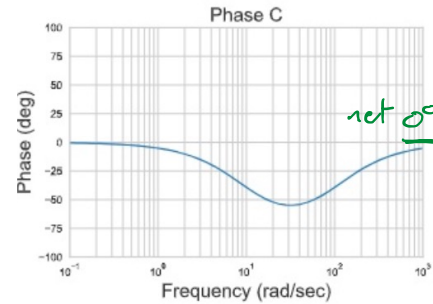
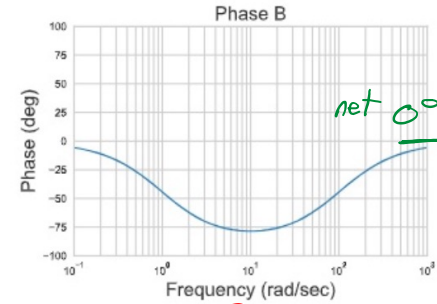
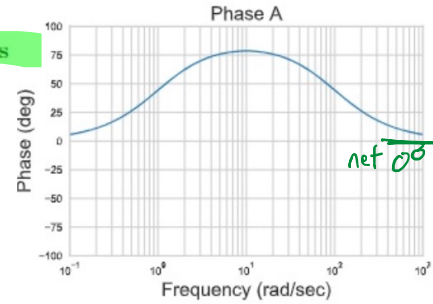


Exam Problem

You are given the following magnitude plot of a plant. Furthermore you know that the plant **does not have any unstable poles.**



Which of the following phase plots corresponds to the above plant?



☐ A Phase plot B

☐ B Phase plot C

☐ C Phase plot D

☐ D Phase plot A

Course Schedule

	Subject	Week
Modeling	Introduction, Control Architectures, Motivation	1
	Modeling, Model examples	2
	System properties, Linearization	3
Analysis	Analysis: Time response, Stability	4
	Transfer functions 1: Definition and properties	5
	Transfer functions 2: Poles and Zeros	6
	Proportional feedback control, Root Locus	7
	Time-Domain specifications, PID control	8
	Frequency response, Bode plots	9
	Polar plots, The Nyquist condition	10
Synthesis	Frequency-domain Specifications, Dynamic Compensation, Loop Shaping	11
	Time delays, Successive loop closure, Nonlinearities	12
	Describing functions	13
	Intro to Uncertainty and Robustness	14

Today

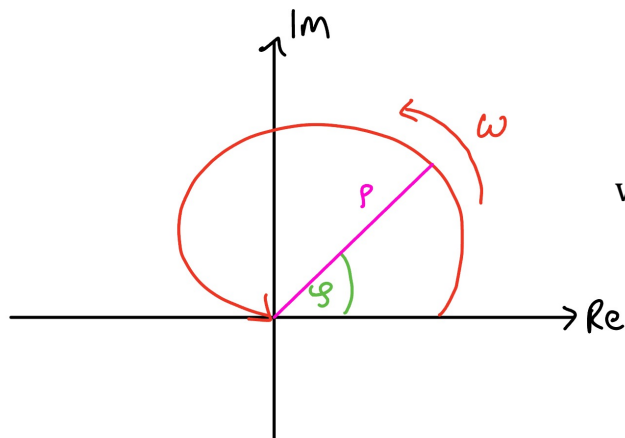
1. Polar Plots
2. Nyquist

1. Polar Plots

Polar Plot

- Instead of showing the magnitude $|G(j\omega)|$ and the phase $\angle G(j\omega)$ on two separate plots, we represent the magnitude & phase for a given ω as a vector in the complex plane:
 - $\rho = |G(j\omega)|$
 - $\varphi = \angle G(j\omega)$
- The frequency is no longer an axis but implicit, just like the gain k in Root Locus

phasor 



we plot $\omega \in [0, +\infty)$

magnitude: $|G(j\omega)| = \frac{10}{\sqrt{\omega^2 + 10^2}}$

Example

phase: $\angle G(j\omega) = -\arctan\left(\frac{\omega}{10}\right)$

Element type	Formally	Magnitude	Phase
single stable pole	$Re(p_i) \leq 0$	$-20 \frac{dB}{dec}$	-90°
single unstable pole	$Re(p_i) > 0$	$-20 \frac{dB}{dec}$	$+90^\circ$
single mnm phs zero	$Re(z_i) \leq 0$	$+20 \frac{dB}{dec}$	$+90^\circ$
single non-mnm phs zero	$Re(z_i) > 0$	$+20 \frac{dB}{dec}$	-90°

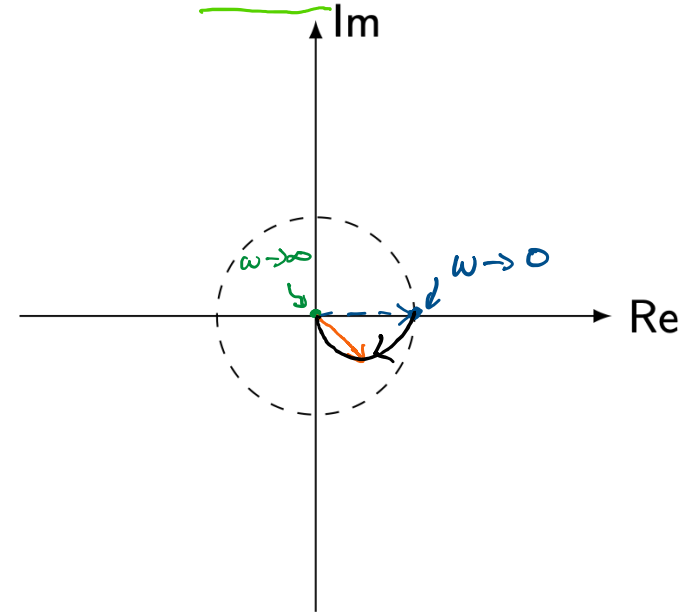
Consider the following TF consisting of a stable pole: $G(s) = \frac{10}{s+10}$

We look at different interesting points:

$G(0) = 1$ $\angle G(0) = 0^\circ$

$|G(\infty)| = 0$ $\angle G(\infty) = -90^\circ$

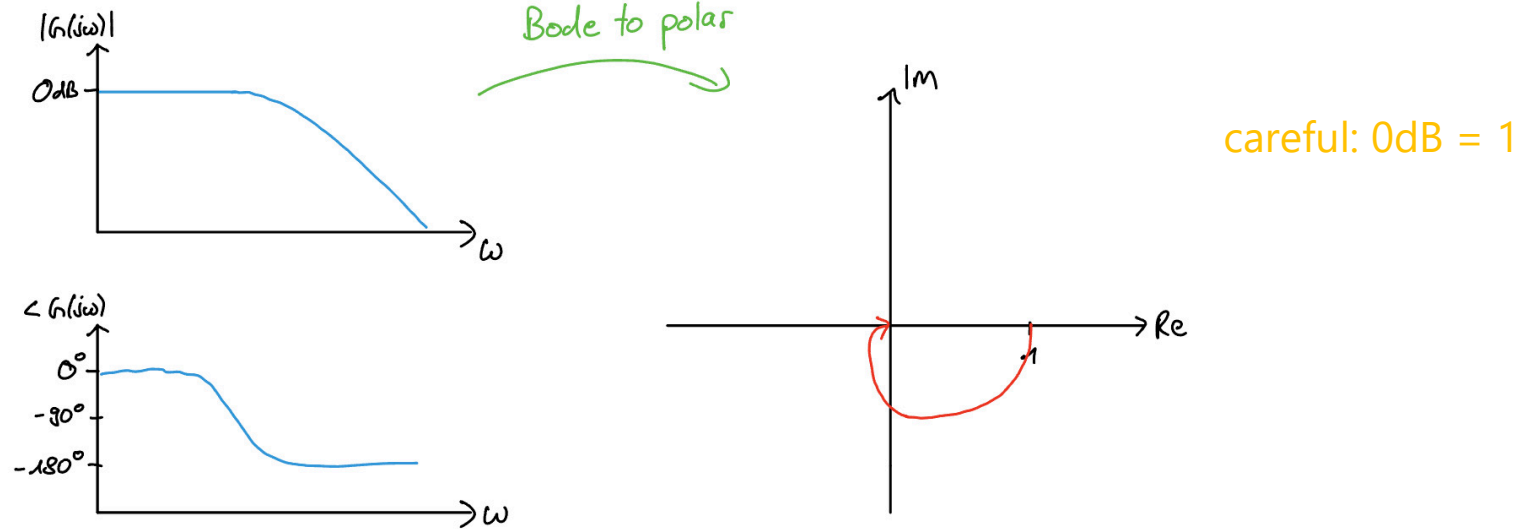
$|G(10i)| = \frac{1}{\sqrt{2}}$ $\angle G(10i) = -45^\circ$



As ω increases, the magnitude goes to 0 and phase to -90°

Bode to Polar Plot

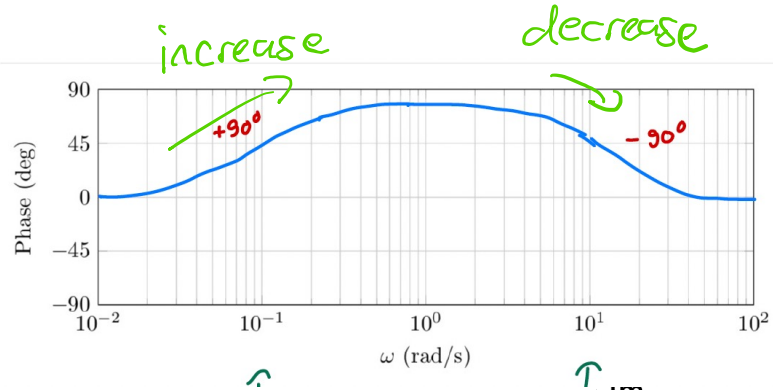
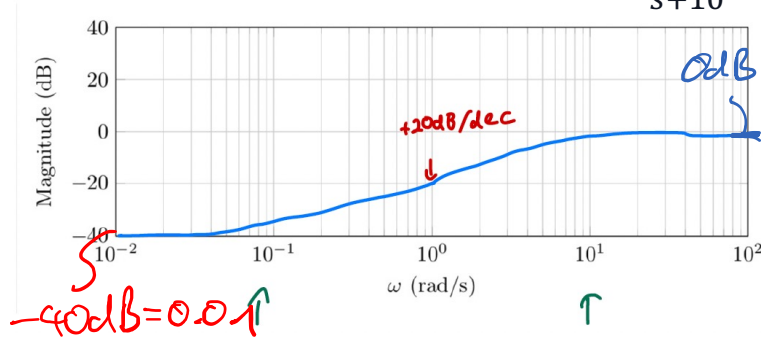
- If we have the Bode plot, we can extract the polar plot as such:



Most often, it will prove simpler to first construct a Bode plot of a system and then derive a polar plot from it

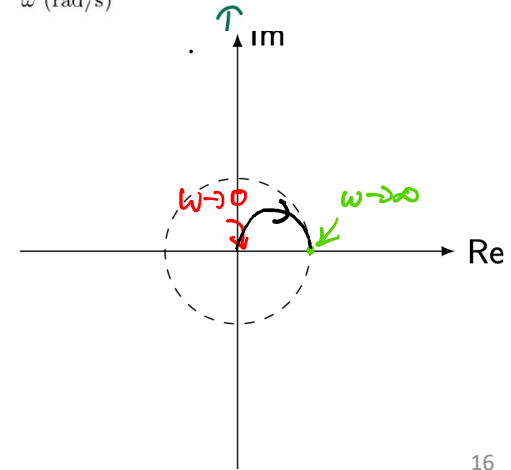
Example

Consider the following TF $G(s) = \frac{s+0.1}{s+10}$



From the Bode plot:

- Magnitude: Increases at first before settling at 0 dB
- Phase: Rises first due to zero and then goes back to 0° due to pole

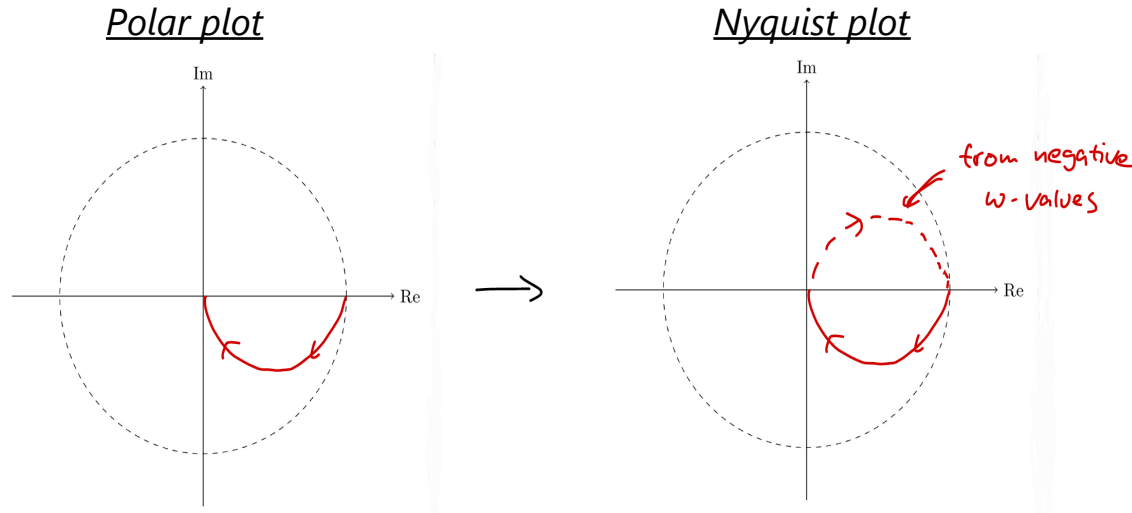


2. Nyquist

Nyquist Plot, Nyquist Stability Criterion

The Nyquist Plot

We can obtain the Nyquist plot by mirroring the polar plot about the real axis
→ this is equivalent to evaluating $G(j\omega)$ for $\omega \in [-\infty, +\infty)$ instead of $\omega \in [0, +\infty)$



Sketch Guide

1. Identify starting point $\omega = 0$
2. Identify end point $\omega = \infty$
(use table from bode plots)
3. Identify intersects with $Re/Im - Axis$
4. Mirror about the $Re - Axis$

Example: $L(s) = \frac{10}{(s+1)(s+2)}$ $L(0) = 5$

1. calculate $L(j0)$ $|L(0)| = 5$ $\angle L(0) = 0^\circ$
2. calculate $L(j\infty)$ $|L(\infty)| = 0$ $\angle L(\infty) = -180^\circ$

3. Intersect with $Im - Axis$:

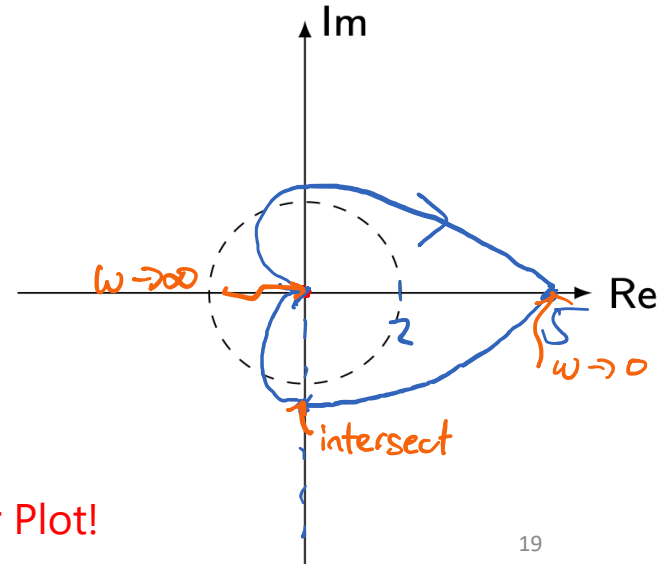
$$L(j\omega) = \frac{10}{(j\omega+1)(j\omega+2)} = \frac{10}{-\omega^2 + 3j\omega + 2} = \frac{10}{2 - \omega^2 + 3j\omega}$$

$$\omega^2 = 2 \Rightarrow \omega = \pm\sqrt{2} \Rightarrow L(\sqrt{2}j) = -2.36j$$

Element type	Formally	Magnitude	Phase
single stable pole	$Re(p_i) \leq 0$	$-20 \frac{dB}{dec}$	-90°
single unstable pole	$Re(p_i) > 0$	$-20 \frac{dB}{dec}$	$+90^\circ$
single mnm phs zero	$Re(z_i) \leq 0$	$+20 \frac{dB}{dec}$	$+90^\circ$
single non-mnm phs zero	$Re(z_i) > 0$	$+20 \frac{dB}{dec}$	-90°

In between: what are the poles/zeros?

- Their order?
 - To the left/right of Im -axis?
- are the mag & phase increasing/decreasing?



For more complex TF draw Bode plot first and then go to Polar Plot!

From TF to Nyquist Plot

Element type	Formally	Magnitude	Phase
single stable pole	$Re(p_i) \leq 0$	$-20 \frac{dB}{dec}$	-90°
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single non-mnm phs zero	$Re(z_i) > 0$	$+20 \frac{dB}{dec}$	-90°

Example:

$$G_1(s) = \frac{20(s+5)}{(s+2)(s-50)}$$

2. $+90^\circ$
1. -90° 3. $+90^\circ$

$$G_2(s) = \frac{20(s-5)}{(s+2)(s+50)}$$

2. -90°
1. -90° 3. -90°

1. find start/end points

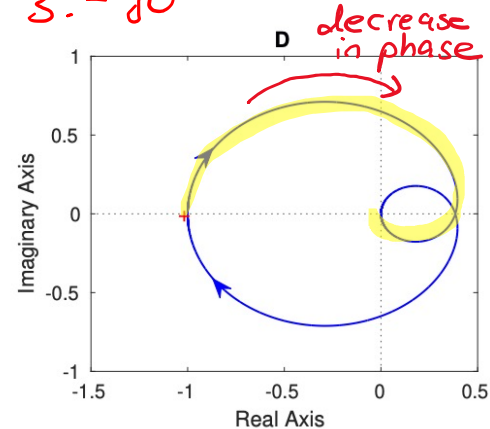
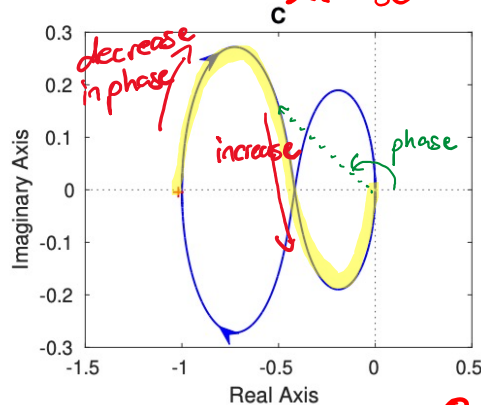
$$G_1(0) = \frac{20 \cdot 5}{-100} = -1 \quad G_1(\infty) = 0$$

$$G_2(0) = -1 \quad G_2(\infty) = 0$$

2. poles/zeros and their phase effects:
look at total phase shift ($\omega \rightarrow +\infty$)

$$G_1 \rightarrow C$$

$$G_2 \rightarrow D$$



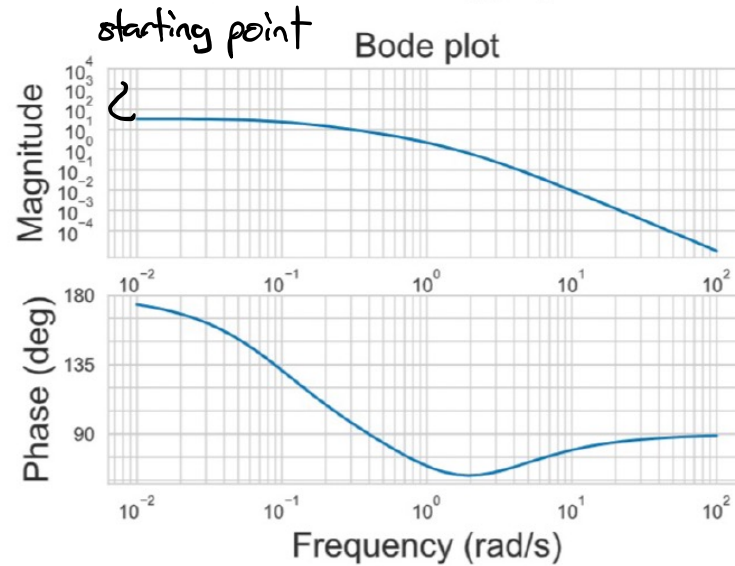
$$\text{net } \angle G_1 = +90^\circ$$

$$\text{net } \angle G_2 = -270^\circ$$

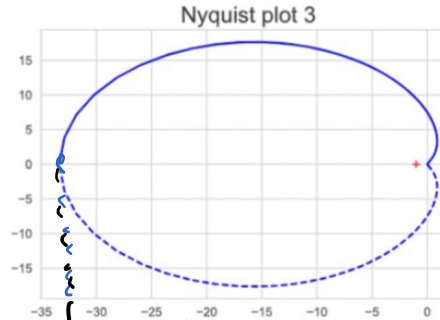
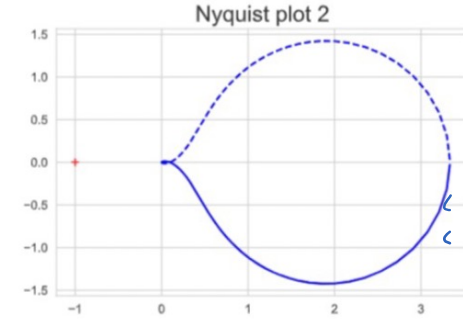
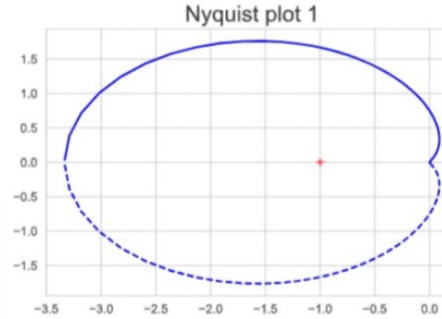
Exam Problem

You are given the following Bode plot.

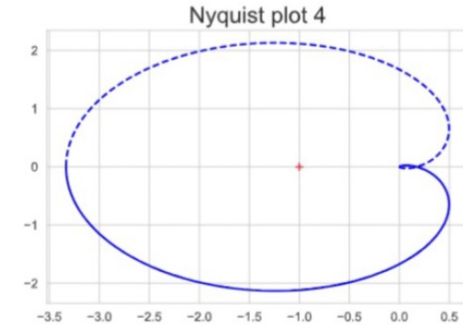
FS 2018



What is the associated Nyquist plot to this Bode plot?



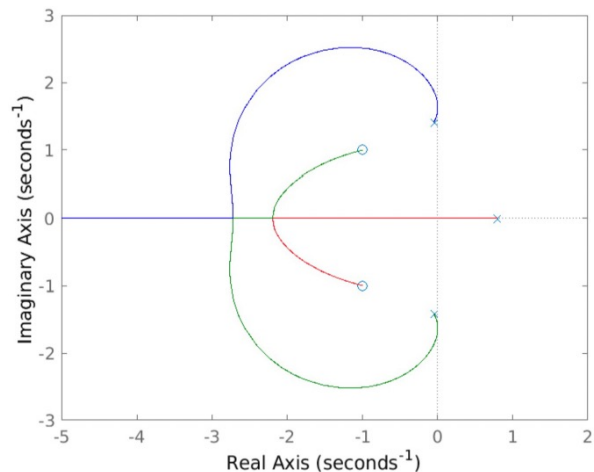
magnitude ≈ 33



Now as we know how to draw Nyquist Plots,
we can move on to...

Nyquist Condition

So far, the tool we have used for assessing stability is the *Root Locus*:



The Root Locus is quicker and simpler, but it has some limitations:

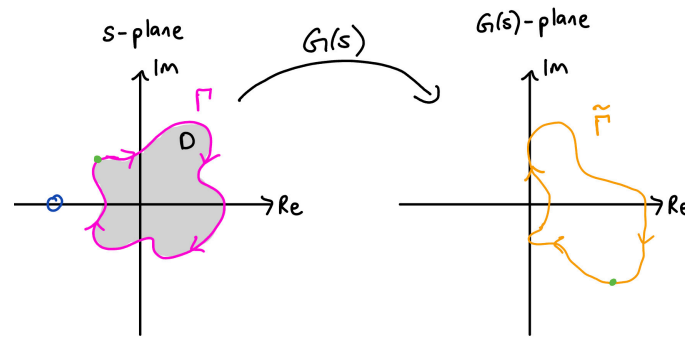
- Can't deal well with time delays (later in the course)
- Struggles with non-rational functions
- Doesn't show gain and phase margins (later today)

→ The Nyquist condition is a more robust method for assessing stability

- [Brian douglas video](#)
- [Chapter 6.4.3 in script](#)

Principle of Variation of Argument

- A little theoretical background for the Nyquist criterion



- D is a region in the s -plane with its boundary Γ
- Evaluating $G(s)$ at every point on the boundary Γ leads to a new boundary $\tilde{\Gamma}$

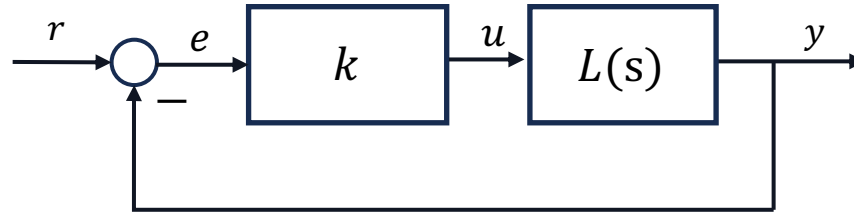
Main Takeaway: When s runs once around Γ , the image $\tilde{\Gamma}$ runs around the *origin* a certain number of times. This number of windings equals the number of zeros minus poles of $G(s)$ located inside the region D

→ **Encirclements of the $\tilde{\Gamma}$ boundary around the origin tell us about the number of poles & zeros in D !**

Encirclements of the $\tilde{r}$ boundary around the origin tell us about the number of poles & zeros in D !

Nyquist Condition

- For a standard feedback loop with proportional control, we derived:



- Closed-Loop Transfer Function $T_{r \rightarrow y} = \frac{kL(s)}{1+kL(s)}$ $\longrightarrow$ poles of CL system: $1 + kL(s) = 0$

Variation
Principle

- VP on open-loop $L(s)$: Encirclements around **origin** relevant
- Closed-Loop: $1 + kL(s) = 0 \longrightarrow L(s) = \frac{-1}{k}$ $\longleftarrow$ **relevant point**

Nyquist Criterion

For stability what interests us are poles with $Re(p_i) > 0$ (unstable poles)

→ for the VP we choose a contour around RHP

The Nyquist Stability Criterion:

$$N = Z - P$$

Z : # unstable closed-loop poles

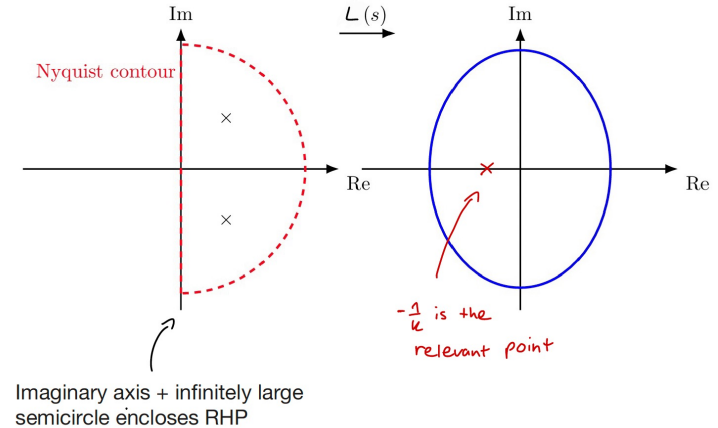
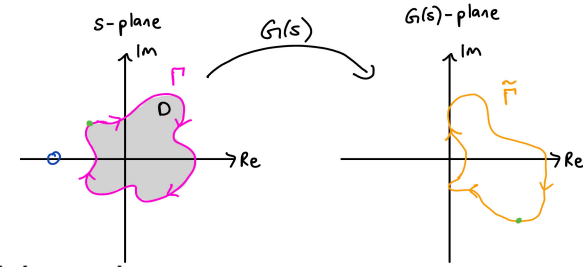
P : # unstable open-loop poles

N : # encirclements around $-1/k$

$N > 0$: CW encirclement (+1)

$N < 0$: CCW encirclement (-1)

For stability: $Z = 0 \rightarrow N + P = 0$



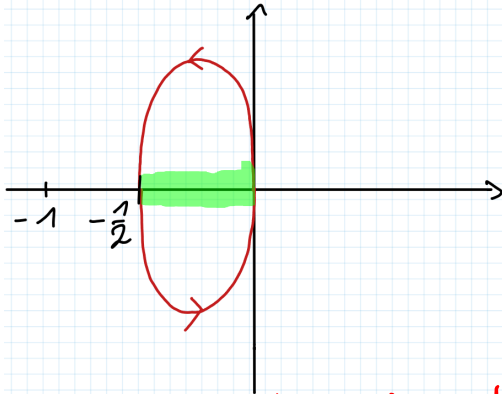
Summarized: We count # of encirclements + # of unstable OL-Poles from $L(s)$

→ can determine if a CL-System is stable or not!
while having only information about the OL-System! 🤖

Example

Given $G(s) = \frac{1}{s-2}$ } 1 unstable open-loop pole
→ unstable system!

Plotting Nyquist of $G(s)$ we get:



1. 1 unstable OL-pole: $P=1$

2. We want stable system $\rightarrow Z=0$
 $\rightarrow N=-1$

The Nyquist Stability Criterion:

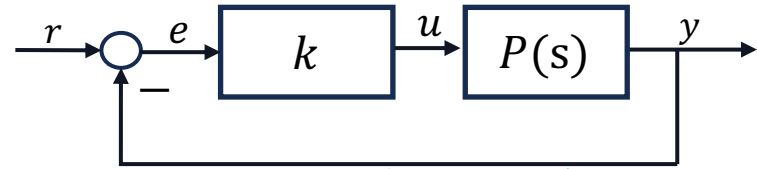
$$N = Z - P$$

Z : # unstable closed-loop poles

P : # unstable open-loop poles

N : # encirclements around $-1/k$

For which values of k can we stabilize the unstable system?



Condition for a stable system

1. $-1/k < 0 \rightarrow k > 0$ ①

2. $-1/k > -1/2 \rightarrow 1/k < 1/2$

$\rightarrow \boxed{k > 2}$ ②

Exam Problem

Problem: Consider the closed-loop system T in Figure 14, where L is a linear time-invariant system with two unstable poles.

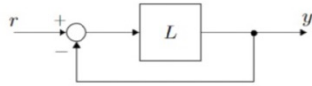


Figure 14: Closed-loop system T .

The Nyquist plot of the transfer function of L is shown in Figure 15.

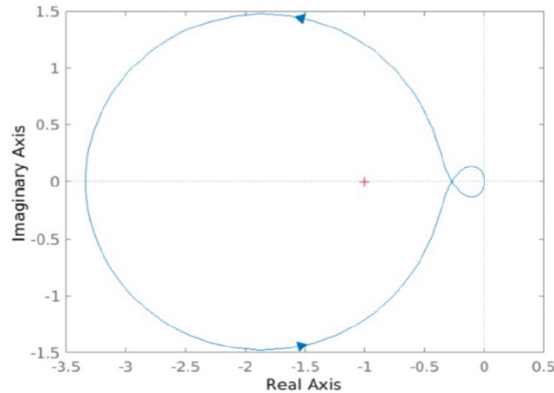


Figure 15: Nyquist plot of the transfer function of L .

Q30 (0.5 Points) Mark the correct answer.

Statement	True	False
The closed-loop system T is stable.	☐	☒

The Nyquist Stability Criterion:

$$N = Z - P$$

Z : # unstable closed-loop poles

P : # unstable open-loop poles

N : # encirclements around $-1/k$

1. $P=2$, $Z=N+P$, we need $Z=0$

$\rightarrow N \stackrel{!}{=} -2$ but there are nowhere 2 ccw encirclements

Poles on imaginary axis

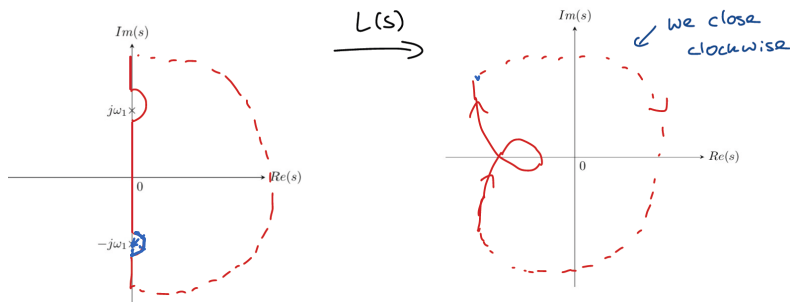
We need to consider the following edge case:

- What if we have p_i with $Re(p_i) = 0$? we can't have any singularities on our contour
- We make small indentions around the poles!

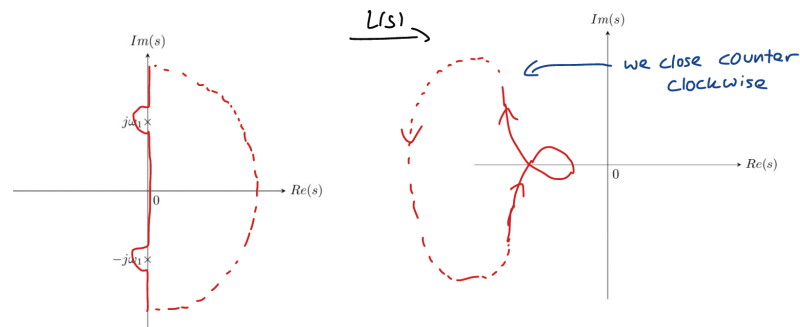
2 ways to close plot:

- If poles are excluded (stable) $\rightarrow$ CW
- If poles are included (instable) $\rightarrow$ CCW

Excluding poles:



Including poles:



Exam Problem

Problem: Consider the feedback system depicted in Figure 16 below.

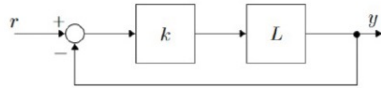


Figure 16: Standard feedback architecture. LTI System L and proportional gain k .

Let the transfer function of the linear time-invariant system L be given by

$$L(s) = 10 \cdot \frac{(s-8)(s-5)}{(s+6)(s+1)(s+10)}$$

The Nyquist plot of $L(s)$ is shown in Figure 17.

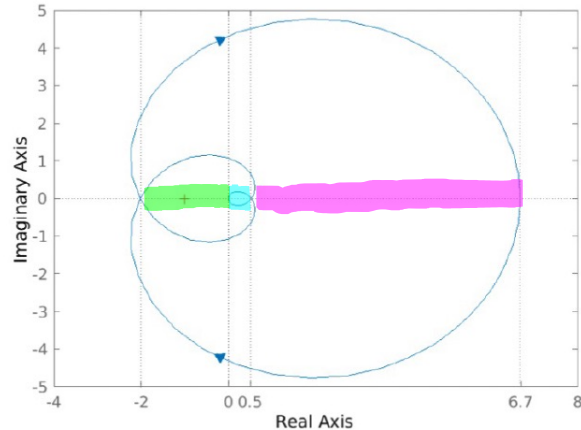


Figure 17: Nyquist plot of $L(s)$.

The Nyquist Stability Criterion:

$$N = Z - P$$

Z : # unstable closed-loop poles

P : # unstable open-loop poles

N : # encirclements around $-1/k$

We now assess the number of closed-loop unstable poles for different values of k .

Q31 (0.5 Points) For $-2 < \frac{-1}{k} < 0$ the number of closed-loop unstable poles Z is

$$Z = 2$$

Q32 (0.5 Points) For $0 < \frac{-1}{k} < 0.5$ the number of closed-loop unstable poles Z is

$$Z = 3$$

Q33 (0.5 Points) For $0.5 < \frac{-1}{k} < 6.7$ the number of closed-loop unstable poles Z is

$$Z = 1$$

Gain & Phase Margin

- These margins are only valid for open-loop stable systems! ($P = 0$) → for stability we need $N = 0$
- Phase & Gain Margin define "how close" we are to closed loop instability (i.e. encircling the -1 point)

1. Gain Margin:

How much can we change the gain at -180° until we hit -1 ?

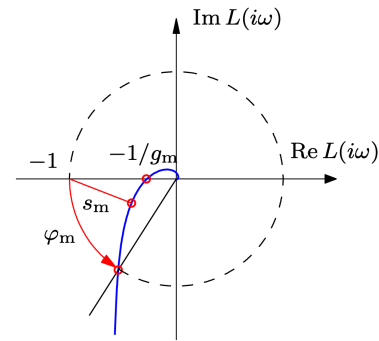
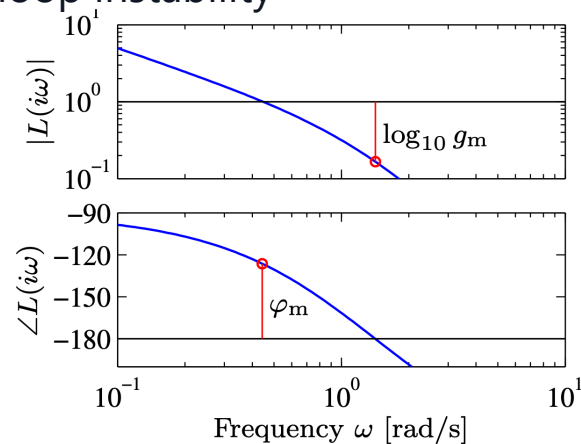
2. Phase Margin:

How much can we change the phase at 0dB until we reach -1 ?

3. Bandwidth:

Frequency where magnitude equals $\frac{1}{\sqrt{2}} = -3\text{dB} = \omega_{gc}$

measures how fast CL-System can react



Exam Problem

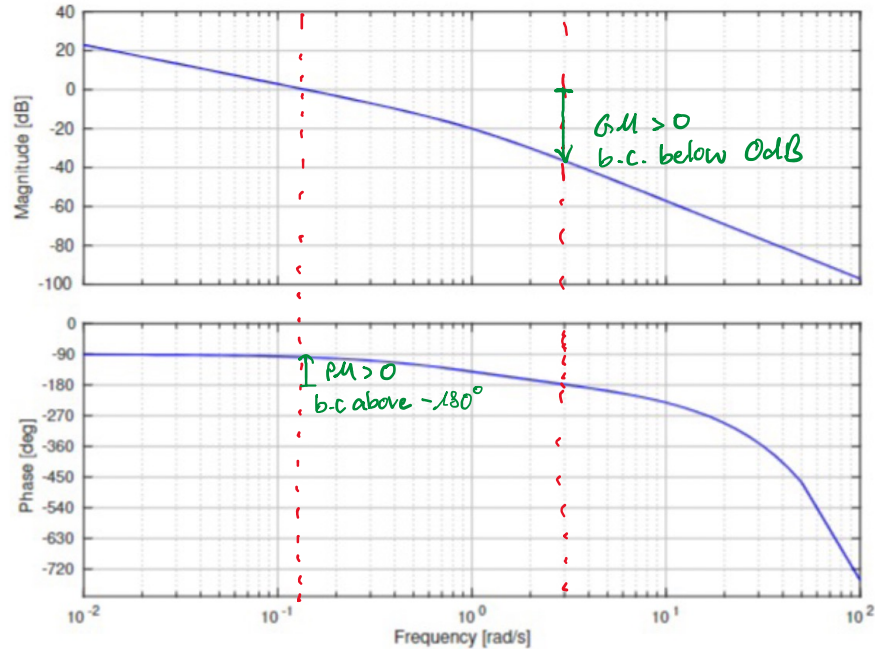


Figure 14: Bode plot of L .

1. Gain Margin:

How much can we change the gain at -180° until we hit -1 ?

2. Phase Margin:

How much can we change the phase at 0dB until we reach -1 ?

Q29 (0.5 Points) Mark the correct answer.

The phase margin PM for closed-loop stability is best described by,

☐ A $PM \approx -83^\circ$

☒ B $PM \approx 80^\circ$

☐ C $PM \approx 47^\circ$

☐ D $PM \approx -35^\circ$

Q30 (0.5 Points) Mark the correct answer.

The gain margin GM for closed-loop stability is best described by,

☐ A $GM \approx -37$ dB

☐ B $GM \approx 82$ dB

☒ C $GM \approx 37$ dB

☐ D $GM \approx -21$ dB

I personally would recommend solving at least the problems marked red

Tips for Problem Set 9

Easy	Medium	Hard
1	2, 3	4

1: Apply the Nyquist stability criterion $Z = N + P$. For each segment values of $-1/k$, determine how many encirclements of the critical point $-1/k$ are required for the closed-loop system to be stable.

2: Focus on identifying key points of interests (i.e $\omega = 0, \omega = \infty$, intersects with Re/Im-Axis). Then apply Nyquist stability criterion.

3: Identify key points of interests (i.e $\omega = 0, \omega = \infty$) look at their magnitude & phase. For the phase as $\omega \rightarrow \infty$ use the tables for the Bode Plot (i.e calculate net phase)

4: Plug in $s = \epsilon e^{j\theta}$ in the TF and look at the phase of the TF as you take the limit $\epsilon \rightarrow 0$. Try not to get lost in algebra, focus on understanding how the phase changes and how this affects the Nyquist closure.

Questions?

Feedback?

Too fast? Too slow? Less theory, more exercises?
I would appreciate your feedback. Please let me know.

<https://docs.google.com/forms/d/e/1FAIpQLSdHI0kjWo63aNzDkAV0cnmQadCAj5L0D7v7aSh0BK7BBdEgpA/viewform?usp=header>